Supplementary Table 1: Search strategy (30.09.2021)

Database		Query	Hits
Medline by PubMed	#1	(((((bronchial asthma[MeSH Terms]) OR (Asthma[Title/Abstract])) OR (wheeze[Title/Abstract])) OR (recurrent wheeze[MeSH Major Topic])) OR ("childhood asthma"[Text Word])) OR ("pediatric asthma"[Text Word])	186852
	#2	((((((adolescent[MeSH Terms]) OR (children[MeSH Terms])) OR (school age population[MeSH Terms])) OR (childhood[Title/Abstract])) OR (children[Title/Abstract])) OR (adolescent*[Title/Abstract])) OR (pediatric*[Title/Abstract])	3703739
	#3	(((((((((calcitriol[MeSH Terms]) OR (cholecalciferol[MeSH Terms])) OR (1,25 dihydroxy 20 epi vitamin d3[MeSH Terms])) OR (vitamin D[Title/Abstract])) OR (cholecalciferol[Title/Abstract])) OR (calcitriol[Title/Abstract])) OR (1,25 dihydroxy 20 epi vitamin d3[Title/Abstract])) OR (vitamin D[Title/Abstract])) OR (25(OH)D[Title/Abstract])) OR ("25 hydroxy D"[Title/Abstract])	84274
	#4	(randomised controlled trial [pt] OR controlled clinical trial [pt] OR randomised [tiab] OR placebo [tiab] OR drug therapy [sh] OR randomly [tiab] OR trial [tiab] OR groups [tiab])	5119779
	#5	**#1 AND #2 AND #3 AND #4**	**225**
Embase	#1	'vitamin d'/exp OR 'colecalciferol derivative'/exp OR 'vitamin deficiency'/exp OR 'vitamin d':ti,ab,kw OR calcitriol:ti,ab,kw OR '25 hydroxyvitamin d':ti,ab,kw OR (25:ti,ab,kw AND oh:ti,ab,kw AND d:ti,ab,kw)	238313
	#2	'child'/exp OR 'pediatric'/exp OR 'adolescent'/exp OR child:ti,ab,kw OR pediatrics:ti,ab,kw OR childhood:ti,ab,kw OR 'school age population':ti,ab,kw	4191782
	#3	'asthma'/exp OR 'recurrent wheezing'/exp OR asthma:ti,ab,kw OR 'reactive airway disease':ti,ab,kw OR wheezing:ti,ab,kw	328446
	#4	'randomized controlled trial'/exp OR 'randomized controlled trial':ti,ab,kw OR randomization:ti,ab,kw OR placebo:ti,ab,kw	917524
	#5	**#1 AND #2 AND #3 AND #4**	**148**
Web of Science	#1	TS=('vitamin D OR 'cholecalciferol derivative' OR 'vitamin D deficiency' OR 'vitamin d' OR calcitriol)	106503
	#2	TS=(child OR pediatric OR adolescent OR paediatric OR childhood OR 'school-age population' OR Child*)	1687630
	#3	TS=(asthma OR 'recurrent wheezing' OR 'reactive airway disease' OR wheezing)	194040
	#4	TS=(randomi*ed controlled trial OR controlled clinical trial OR randomized OR placebo OR randomly OR trial OR groups)	5807787
	#5	**#1 AND #2 AND #3 AND #4**	**437**
CENTRAL	#1	MeSH descriptor: [Asthma] explode all trees	11796
	#2	("asthmatic"):ti,ab,kw	8496
	#3	#1 OR #2	15698
	#4	MeSH descriptor: [Vitamin D] explode all trees	5541
	#5	("vitamin D"):ti,ab,kw	12448
	#6	("cholecalciferol"):ti,ab,kw	3009
	#7	#4 OR #5 OR #6	13544
	#8	#3 AND #7	140
	#9	MeSH descriptor: [Child] explode all trees	56872

	#10	(children):ti,ab,kw	150873
	#11	("adolescent"):ti,ab,kw	130812
	#12	("school age"):ti,ab,kw	1335
	#13	#9 OR #10 OR #11 OR #12	237337
	#14	**#8 AND #13**	**86**

Supplementary Table 2: Definition of Asthma attack used in different studies.

Author (Year)	Definition of asthma attacks
Urashima M et al. (2010)[1]	Wheezing improved by inhalation of a β stimulant in patients who already had a diagnosis of asthma
Majak P et al. (2011)[2]	Not mentioned.
Yadav M et al. (2013)[3]	Not mentioned.
Jensen ME et al. (2016)[4]	Asthma attacks require rescue oral corticosteroids (documented in medical and/or pharmacy records).
Kerley CP et al. (2016)[5]	They have not mentioned in the primary study. However, they provided data on asthma attacks requiring oral corticosteroids (mentioned in individual patient data meta-analysis by Jolliffe et al.).
Tachimoto H et al. (2016)[6]	Not mentioned. However, provided data on asthma attacks requiring systemic corticosteroids
Alansari K et al. (2017)[7]	Requiring an unplanned visit for asthma, recommended by a joint expert committee
Ducharme FM et al. (2019)[8]	Asthma attacks requiring oral corticosteroids
Forno E et al. (2020)[9]	Mentioned severe asthma attacks defined as the occurrence of either (1) use of systemic corticosteroids (tablets, suspension, or injection) for at least 3 days or (2) a hospitalization or ED visit because of asthma, requiring systemic corticosteroids.
Jat KR et al. (2021)[10]	Any asthma attack requires rescue medications (beta-agonist or corticosteroids) or an emergency visit (Personal communication).
Thakur C et al. (2021)[11]	Asthma requiring oral corticosteroids (personal communication, not mentioned in manuscript).

Supplementary Table 3: Random-effect Meta-regression Analysis for Co-Variates (Sample Size, Cumulative dose of vitamin D, Active Control, i.e., vitamin D in the control group, and co-treatment)

Log Risk Ratio	Coefficient	Standard Error (95% CI)	p-value
Use of Rescue Systemic Corticosteroids (6 studies)			
Sample Size	0.0011957	0.0194735 (- 0.2462385 to +0.2486299)	0.96
Cumulative Dose	2.82×10^{-7}	2.08×10^{-6} (-2.62×10^{-5} to $+2.68 \times 10^{-5}$)	0.91
Active Control use	-0.7138567	0.7290598 (-9.977439 to + 8.549726)	0.51
Co-treatment	0.1681983	0.5177693 (-6.410685 to + 6.747081)	0.80
Constant	-0.0289788	0.4642203 (-5.927457 to + 5.869499)	0.96
One or more asthma attacks of any severity (11 studies)			
Sample Size	0.0000952	0.0024565 (- 0.0059157 to +0.006106)	0.97
Cumulative Dose	4.65×10^{-7}	5.01×10^{-7} (-7.61×10^{-7} to $+1.69 \times 10^{-7}$)	0.39
Active Control use	-0.2410449	0.3767945 (-1.163028 to + 0.6809379)	0.55
Co-treatment	0.0103854	0.4297825 (-1.041255 to + 1.062025)	0.98
Constant	-0.2675291	0.3583356 (-1.144345 to + 0.6092866)	0.48

Supplementary Table 4: Comparison of Adverse Events

Outcome	No. of studies (Participants)	Vitamin D (n/N)	Control (n/N)	RR [95% CI]	I^2 p-value
Any Adverse Event					
Nausea	3 (503)	10/252	8/251	1.21 [0.50, 2.94]	0%, 0.8
Vomiting	1 (250)	34/125	28/125	1.21 [0.79, 1.87]	Not Applicable
Pain Abdomen	1 (250)	40/125	41/125	0.98 [0.68, 1.40]	Not Applicable
Constipation	1 (250)	12/125	11/125	1.09 [0.50, 2.38]	Not Applicable
Headache	1 (250)	25/125	25/125	1.00 [0.61, 1.64]	Not Applicable
Seizures	1 (250)	1/125	0/125	3.00 [0.12, 72.94]	Not Applicable
Altered Sensorium	1 (250)	0/125	1/125	0.33 [0.01, 8.10]	Not Applicable
Rash	1 (231)	1/116	0/115	2.97 [0.12, 72.26]	Not Applicable
Serious Adverse Events					
All*	3 (525)	11/271	8/254	1.30 [0.55, 3.07]	0%, 0.9
Hospitalization	2 (275)	9/146	6/129	1.38[0.52, 3.66]	0%, 0.8

Abbreviations: **CI:** Confidence interval; **RR:** Risk ratio

*Includes hospitalization.

n/N represents the number of events/ total number of participants.

Supplementary Table 5: Sensitivity Analysis (Low risk of Bias studies)

Outcome	No. of studies (Participants)	RR/MD [95% CI]	Heterogeneity (I^2), p-value
Participants requiring rescue systemic corticosteroids	6 (445)	1.13 [0.86, 1.48]	0%, 0.7
Participants with ≥ 1 asthma exacerbation	8 (702)	0.97 [0.75, 1.26]	19%, 0.3
Unscheduled healthcare visits	2 (130)	1.58 [0.91, 2.74]	0%, 0.9
Well-controlled asthma	3 (342)	1.01 [0.96, 1.07]	0%, 0.9
Hospitalization	2 (275)	1.05 [0.45, 2.45]	0%, 0.7
Serious Adverse events	3 (525)	1.30 [0.55, 3.07]	0%, 0.9
Serum 25(OH) D levels	7 (626)	10.77 [5.44, 16.10]	94%, 0.01
FEV1	4 (314)	-2.64 [-7.04, 1.77]	62%, 0.05
FeNO	2 (94)	-2.87 [-24.66, 18.91]	75%, 0.05

Abbreviations: CI: Confidence interval; MD: Mean difference; RR: Risk ratio

Explanations: [a] 95% CI crosses' null line; [b] Extremely wide 95% CI; [c] High heterogeneity.

Supplementary Table 6: Sensitivity Analysis (Fixed effect)

Outcome	No. of studies (Participants)	RR/MD [95% CI]	Heterogeneity (I^2), p-value
Participants with ≥ 1 asthma exacerbation requiring systematic corticosteroids	6 (445)	1.09 [0.83, 1.43]	0%, 0.7
Unscheduled healthcare visits	3 (361)	1.00 [0.90, 1.11]	0%, 0.4
Well-controlled asthma	4 (442)	1.01 [0.96, 1.06]	0%, 0.9
Hospitalization for asthma exacerbations	2 (275)	1.06 [0.46, 2.47]	0%, 0.7
Serious Adverse events	3 (525)	1.30 [0.55, 3.08]	0%, 0.9

Abbreviations: CI: Confidence interval; MD: Mean difference; RR: Risk ratio